IEEE Access
Multidisciplinary ⋮ Rapid Review ⋮ Open Access Journal

Received 21 August 2024, accepted 5 September 2024, date of publication 19 September 2024, date of current version 30 September 2024.

Digital Object Identifier 10.1109/ACCESS.2024.3463793

RESEARCH ARTICLE

A Deep Learning Model for Student Sentiment Analysis on Course Reviews

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This work was supported in part by the Natural Science Foundation of the Higher Education Institutions of Anhui Province under Grant KJ2021A0484, in part by Anhui Province Philosophy and Social Sciences Planning Project under Grant AHSKF2021D31, and in part by the Science Research Foundation of Anhui University of Finance and Economics under Grant 90084.

ABSTRACT The learning status of students directly influences their learning outcomes. Course evaluations by students offer valuable insights into the effectiveness of teaching methodologies and students' own perceptions of their learning environments. These evaluations encapsulate a spectrum of emotions, ranging from satisfaction to frustration, and provide a holistic view of the educational experience. Analyzing course evaluations plays a crucial role in deciphering students' learning outcomes. By extracting sentiments from these evaluations, educators can gain a deeper understanding of students' experiences, identifying areas of improvement and tailoring teaching strategies to better meet students' needs. Sentiment analysis of course evaluations falls under text sentiment analysis. In this study, we leverage advanced deep learning techniques to extract sentiment from course evaluations. Specifically, we capitalize on the capabilities of BERT (Bidirectional Encoder Representations from Transformers), a state-of-the-art language model known for its proficiency in understanding context and semantic meaning. By harnessing BERT's bidirectional encoding and feature extraction capabilities, coupled with the bidirectional influence of BiLSTM (Bidirectional Long Short-Term Memory) models, we aim to capture nuanced relationships within the text data. Moreover, we incorporate the attention mechanism into our model architecture to assign varying weights to different parts of the input sequence, enabling the model to focus on relevant information effectively. The proposed BERT-BiLSTM-ATTENTION (BBA) model represents an innovative approach that capitalizes on the synergy between these advanced techniques. Our experimental results, based on real-world data extracted from online comments on Massive Open Online Courses (MOOCs), demonstrate the superior performance of the BBA model compared to existing approaches. By thoroughly exploring bidirectional semantics and accounting for both preceding and subsequent influences, our model offers enhanced predictive capabilities and deeper insights into students' learning experiences. This has significant implications for enhancing students' learning outcomes and improving teaching methods.

INDEX TERMS Course comments, sentiment analysis, BERT, BiLSTM, attention mechanism, personalized teaching.

I. INTRODUCTION

The learning status directly affects the effectiveness of learning, and timely acquisition of students' learning status is of paramount importance for ensuring prompt mastery and adjustment of their learning trajectories. Analyzing the emotional nuances embedded in student learning experiences provides a valuable avenue for gaining insights into their learning status. Sentiment analysis of students' learning status has emerged as a prominent area of research in education, directly influencing the effectiveness of student learning and individual growth.

The associate editor coordinating the review of this manuscript and approving it for publication was Jolanta Mizera-Pietraszko.

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The proliferation of online learning platforms has resulted in a vast accumulation of textual data, including students' comments on course materials, teaching experiences, and course evaluations. These comments often reflect learners' attitudes toward learning, their cognitive states, and their satisfaction with teaching quality and content, all imbued with strong emotional nuances. This offers a promising opportunity to leverage course evaluation data for sentiment analysis.

Different emotional states can yield varying learning outcomes. Positive emotions can enhance learners' focus and motivation, while negative emotions may hinder their enthusiasm for learning and subsequently impact their learning outcomes. Identifying and addressing negative emotions in a timely and targeted manner can facilitate a transition to a more positive emotional state, thereby enhancing learning outcomes. Sentiment analysis enables the swift detection of students' potential confusion, dissatisfaction, or frustration, allowing educators to identify course-related problems or issues with teaching methods early on and make necessary adjustments to enhance the overall learning experience, including teaching methods, content, pace, and strategies.

Therefore, accurate analysis and discerning students' emotions are crucial for gauging their learning status and effectiveness. Exploring emotional tendencies within learning review data, analyzing students' emotions, and conducting sentiment analysis of their learning status not only represent an inevitable trend in educational research but also serve as vital approaches to improving teaching quality and nurturing student development.

Commentary text data plays a crucial role in online learning environments, holding significant importance for understanding student feedback, improving teaching quality, and providing personalized support. However, effectively leveraging this data requires integrating sentiment analysis techniques to overcome challenges related to data quality and emotional diversity. These challenges include the complexity of semantic understanding, considerations of context, and the subjectivity of emotions.

There are no practical methods to address problems such as the brevity and sparsity of comments, the open-ended range of vocabulary, and occasional word order inversion [1], [2]. Comments may contain grammar errors, be short in length, include internet slang, lack context, etc., all of which increase the complexity and accuracy challenges of sentiment analysis. Currently, teaching evaluation data in online learning predominantly consists of short-text information. Short-text data is difficult to analyze due to its sparsity [3]. Therefore, sentiment analysis based on online comment texts still faces significant challenges.

Pre-trained knowledge can enrich the semantics of the text, while capturing bidirectional sequence dependencies within the text helps fully extract meaning, thereby mitigating the impact of short text and data sparsity. The differentiated weighting of information importance allows for better focus on crucial information, thereby improving the quality of feature representation. This paper proposed a BERT-BiLSTM-ATTENTION (abbreviated as BBA) network architecture. The model can better extract text features and improve the accuracy of sentiment analysis.

The main contributions of this paper are as follows:

1) We introduced a new approach to sentiment analysis using BERT, BiLSTM and the attention mechanism. The combination of BERT, which excels at capturing deep-level information, BiLSTM, which effectively captures bidirectional sequence dependencies, and the attention mechanism, which focuses on the most important information, can further enhance the extraction of temporal features from text.

2) We collected real-world course review data and conducted experiments on it. This real-world data enhances the proposed model's application in the education field, offering better generalization compared to controlled laboratory data and better supporting the development of intelligent education.

3) We conducted substantial experiments from various perspectives. We compared the model performance with that of several state-of-the-art sentiment analysis models, as well as evaluated the effects of the attention mechanism and the bidirectionality of BiLSTM on model performance. The results showed that our model was superior.

The structure of this paper is organized as follows. In Section II, we introduce the related work in sentiment analysis, the existing methods and models. Section III presents the model proposed in detail and related mechanism. Our experimental process and performance comparison are done on different models in Section IV. We give a conclusion of the whole paper in Section V.

II. RELATED WORK

A. RELATED TECHNOLOGIES OF SENTIMENT ANALYSIS

Sentiment analysis is an important branch in natural language processing. It is a method of recognizing, extracting, and quantifying emotional colors in text through natural language processing techniques. Text sentiment analysis is the most common type of sentiment analysis. According to different classification standards, sentiment analysis can be divided into different categories.

If divided according to the granularity of analysis, it can be divided into document-level sentiment analysis [4], sentence-level sentiment analysis [5], topic-based sentiment analysis, and aspect-based sentiment analysis [6], [7]. Document-level sentiment analysis focuses on the overall sentiment orientation of the entire document, typically used to analyze sentiments in texts such as news articles, product reviews, etc. Sentence-level sentiment analysis, on the other hand, is a method of analyzing sentiments for individual sentences or phrases, focusing on the sentiment orientation of each sentence. It is commonly employed to analyze sentences or paragraphs within lengthy texts to accurately capture changes in sentiment. Topic-based sentiment analysis

involves analyzing sentiments related to specific topics or themes within a text. It focuses on sentiments relevant to particular topics, often applied to analyze text data from social media comments, forum posts, etc., to understand people's opinions and sentiments regarding specific topics. Aspect-based sentiment analysis is a method of analyzing sentiments related to specific aspects or attributes within a text. It examines sentiments associated with different aspects or attributes within the text, commonly used to analyze text data such as product reviews, restaurant evaluations, etc., to understand people's opinions and sentiments regarding specific aspects or attributes, such as the appearance, performance, price, etc., of a product.

From a technical perspective, sentiment analysis primarily involves methods based on lexicons, machine learning, deep learning, and hybrid approaches.

Lexicon based approaches: This method uses a sentiment lexicon or vocabulary to calculate the number and polarity of sentiment words in the text by matching the words in the text with the entries in the sentiment lexicon. Based on this information, the sentiment tendency of the text is obtained. Many scholars have constructed sentiment lexicon and conducted sentiment analysis based on the created sentiment lexicon, achieving good results.

Ojeda-Hernández et al. [8] proposed using Formal Concept Analysis (FCA) to create classification lexicon. This method was different from methods that relied on predefined lexicons, allowing for the creation of customized lexicon for specific data and tasks. The experiment showed that this method achieved better performance than other standard lexicons. Khoo and Johnkhan [9] proposed a WKWSCI universal sentiment lexicon and compared its sentiment classification with five other sentiment lexicons on different tasks. Experimental results had shown that this lexicon performed better than other species lexicons in sentiment classification of news headlines. Xu et al. [10] and Zhang et al. [11] extended the sentiment lexicon and designed sentiment scoring rules and sentiment score weight calculation based on the extended sentiment lexicon. Reference [10] utilized the extended sentiment lexicon and the designed sentiment score rules to achieve the sentiment of the text. Reference [11] extended the sentiment lexicon and computed the weight for the sentiment value.

Lexicon-based sentiment analysis heavily relies on the quality of sentiment lexicons. If the quality of the lexicon is not good, the classification performance will be greatly reduced. There is an emotional deviation problem [12].

Machine learning approaches: using supervised or unsupervised learning techniques to train models to recognize emotions in text. Supervised learning methods require labeled datasets as training samples, including text and corresponding emotional labels. Unsupervised learning methods discover emotional patterns by clustering or pattern recognition of text, without the need for prelabeled training data. Bayesian, KNN, and SVM et al. are traditional machine learning methods commonly used in sentiment analysis.

Obiedat et al. [13] handled sentiment analysis of customers' reviews by combining the Support Vector Machine (SVM) algorithm with Particle Swarm Optimization (PSO), which produced the best results compared to different classification techniques in terms of multiple criteria. Benarafa et al. [14] experimented with the proposed K-Nearest Neighbors (KNN) based method for implicit aspect identification on two datasets, electronic products and restaurant reviews, improving performance in handling overfitting and underfitting problems.

Rehioui and Idrissi [15] combined two well-known clustering methods: K-means and DENCLUE with its variants to analyze sentiment of twitter. The algorithm was better than state-of-the-art methods to provide a tradeoff between clustering performance, the number of returned clusters, and runtime.

Sentiment classification based on supervised learning has a wide range of applications, so text sentiment analysis based on machine learning relies on sufficient labeled training data [16]. Natural semantic ambiguity and indirect sentiments also make it difficult to use traditional machine learning methods for sentiment classification. On the other hand, machine learning based methods often require manual selection and extraction of important features from input data, which is a laborious process that requires domain knowledge.

Deep learning has the advantage of automatically learning data features and is currently widely used.

Deep learning methods: These are approaches based on neural networks. Deep learning methods can more accurately capture emotional semantics and emotional changes by learning complex patterns and contextual information in text sequences, making them suitable for processing larger and more complex data. Due to the advantages of deep learning, many scholars use it for sentiment analysis.

Convolutional Neural Networks (CNNs) are widely employed and known for their effectiveness. By training deeper network models, CNNs enhance generalization abilities, rendering them extensively utilized in various applications. Su et al. [17] improved the accuracy of sentiment analysis by combining CNN with emotions and attention mechanisms. Luo [18] proposed a GRU-CNN model based on LDA text representation to improve the performance of sentiment analysis. A combination of Latent Dirichlet Allocation (LDA) text representation and CNN is proposed. Gao et al. [19] performed sentiment analysis on short texts based on CNN and BiGRU, and calculated feature vector weights using TF-IDF. Yan et al. [20] proposed a parallel DenseNet to realize sentiment analysis which solved the problem of ignoring global features in traditional CNN models.

Due to the use of time series in RNN, scholars have utilized this feature for sentiment analysis, which has improved the analysis performance. Huan et al. [21] used RNN to achieve

a higher accuracy which preserved the positional relationship among words.

LSTM solves the long-term dependency problem of RNN and has a wider range of applications in sentiment analysis. He et al. [22] proposed a bidirectional LSTM with a trapezoidal structure, which used multi-layer LSTM to better extract high-dimensional features of text, while the trapezoidal structure reduced the model parameters.

Hybrid Approaches: Some researchers integrate different methods and techniques mentioned above, such as merging lexicon with machine learning or incorporating lexicon into deep learning approaches.

Liu et al. [23] combined rule-based sentiment lexicon and machine learning to design a weight distributing method for text sentiment analysis. It can both highlight words with sentiment meanings while retaining their text information.

Hao et al. [24] proposed a BERT-SVM model whose effect was better than other deep learning methods and machine learning methods. Kamyab et al. [25] fused CNN and BiLSTM neural networks for sentiment analysis. After the rise of attention mechanisms, scholars have integrated them into models and achieved good results. Shin et al. [26] integrated lexicon embeddings and an attention mechanism into convolutional neural networks for sentiment analysis and the attention mechanism effectively dimmed out noisy words. Khan et al. [27] proposed a mixed emotion and context-aware DNN model with attention mechanism, which can intelligently learn and highlight prominent features of relevant emotional contexts in text. Li et al. [28] incorporated an attention mechanism, BiLSTM, and CNN for sentiment analysis which demonstrated a superior performance on a public e-commerce dataset.

B. SENTIMENT ANALYSIS IN EDUCATION

In the sentiment analysis task of educational data mining, scholars have also explored various technical methods. Analyzing the implicit emotions and satisfaction in course comments is an effective way to improve teaching quality. In [29], a method of constructing a sentiment lexicon for online course reviews was proposed, which based on multi-source combination. Scholars have utilized traditional machine learning algorithms for sentiment mining in the field of education. Wang and Zhang [30] highlighted the identification and visualization of topic sentiment based on learning topic mining and sentiment clustering at various granularity levels that outperformed other state-of-art models. Yuan [31] combined PMI and SVM to analyze the sentiment of online legal course and get good effect and can be applied to practice. CNN, LSTM and other deep learning networks are also widely used in the field of educational data mining. The empirical analysis of Onan [32] showed that in the sentiment analysis task of educational data mining, the architecture based on deep learning was superior to ensemble learning methods and supervised learning techniques. Li et al. [33] used the BERT-CNN model as a comment classifier to analyze emotional information in student course comments through deep learning, in order to discover problems and improve teaching effectiveness. However, since CNNs usually perform convolution operations within a local window, this may lead to the loss of some global context information. Due to BiLSTM's ability to better capture bidirectional semantic relationships, so it is widely used in emotional mining in education. Qi and Liu [34] used LDA to mine the course reviews and obtain a topic-word distribution matrix and a comment-topic distribution matrix. Next, calculated the emotion values of the comments in each topic and proposed a BiLSTM text classification model. However, since LDA is based on the bag-of-words assumption, it ignores word order and contextual information, which may lead to inaccurate topic modeling and, consequently, affect the effectiveness of BiLSTM applications. Shen and Tan [35] applied BiLSTM for sentiment analysis of course comments, introducing the simple copy method and Easy Data Augmentation Techniques (EDA) to balance the training dataset, and adding a dropout mechanism to prevent model overfitting. Due to its relatively simple structure, it is not strong enough in handling complex contexts. Scholars have also integrated two or more types of deep learning networks. Zhang and Qin [36] integrated CNN and LSTM to identify and analyze the emotional tendency of comments on online ideological and political courses. The model is unable to effectively capture contextual information. Wang et al. [37] proposed a hybrid neural network model (BiAC-HNN), which used BiLSTM to obtain contextual semantic information of sentences, combined attention to obtain feature vectors with attention scores, and then used CNN to obtain higher-level key features. The accuracy and F1 value of the sentiment orientation of the course text obtained by this model were better than BiLSTM, CNN, and their enhanced models. But the model is complex and difficult to tune. Table 1 compares the applications of deep learning in the field of course evaluation.

The key to text sentiment classification is to determine the features and corresponding weights that recognize emotions in the text. Context, ambiguity, satire, and inferential semantics are difficulties in handling sentiment classification [38]. To further make full use of the information in comment text and enhance sentiment classification results, this paper presents a deep learning-based model, using the advantages of BERT and BiLSTM, joining the attention mechanism, and giving different values according to the impact of different words on the results. The model performs excellently in sentiment classification.

III. METHODOLOGY

A. BERT-BiLSTM-ATTENTION MODEL FRAMEWORK

The BBA model consists of seven layers: an input layer, a BERT layer, a BiLSTM layer, an Attention layer, a Pooling layer, a fully connected layer and an output layer, which is based on BERT and attention mechanism. The framework structure of the model is illustrated in Fig. 1.

TABLE 1. Comparison of related studies.

Author	Models	Evaluation index	Extra feature
Li[33]	BERT-CNN	Accuracy,F1	May cause some global context information to be lost
Qi[34]	LDA, BiLSTM	Accuracy	Ignoring word order of LDA can weaken the effectiveness of BiLSTM
Shen[35]	BiLSTM	Accuracy,Recall,F1	Simple in structure , not strong enough in handling complex contexts
Zhang[36]	CNN-LSTM	Precision, Recall, F1	Unable to effectively capture contextual information
Wang[37]	BiLSTM-Attention-CNN	Accuracy,Recall,F1	Model is complex and difficult to tune

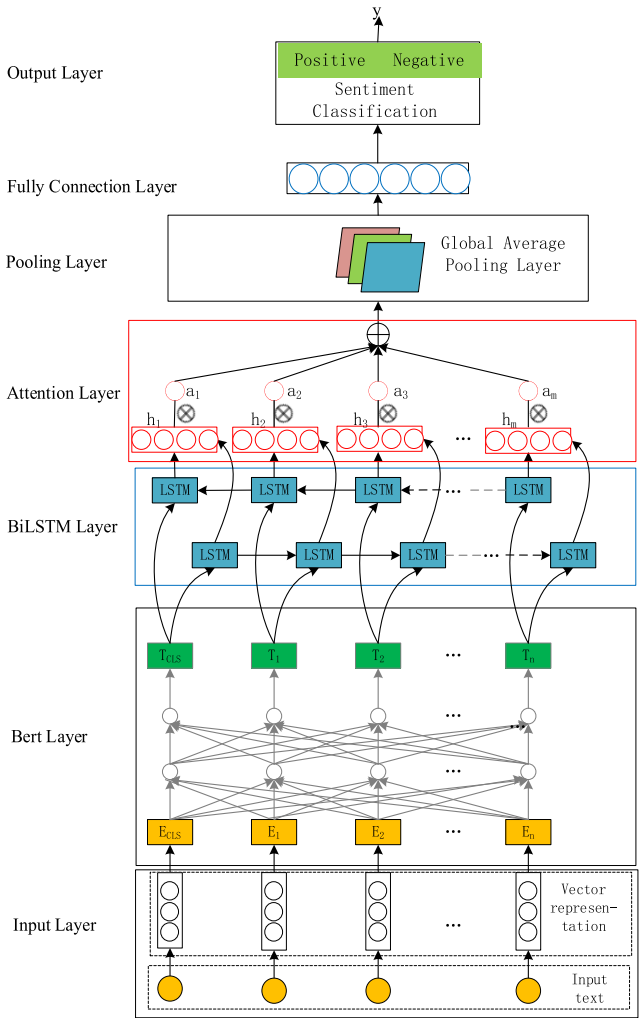

FIGURE 1. BBA model framework.

1) INPUT LAYER

Preprocess the course review texts, including removing erroneous symbols and other confusing characters, encoding the review texts into numerical representations to obtain the final input data.

2) BERT LAYER

Feed the word embedding vector into the Bert, which is a stack of transformer encodings that can bi-directional encode text and generate contextual representations for each word.

3) BILSTM LAYER

The output vector of BERT is fed into bidirectional LSTM, respectively, in a different order, so BiLSTM can further captures sequence information and contextual information, and produces more representative representations, enabling a better understanding of word dependencies.

4) ATTENTION LAYER

It can enable the model to focus on the most critical part of sentiment classification in the text. By using attention mechanisms, models can assign different weights to various parts of text information, enhancing important contextual information and reducing reliance on irrelevant details. It can improve the model's understanding and processing capabilities of text sentiment.

5) POOLING LAYER

Global Average Pooling can average pool vector obtained by the attention layer, reducing its dimensionality into a fixed-length vector, thus reducing the number of parameters in subsequent layers and lowering computational complexity. Simultaneously, it can aggregate information from the entire sequence, extracting global features of the sequence, enabling the model to comprehensively understand the entire input sequence.

6) FULLY CONNECTED LAYER

Input the features to the classification layer, perform nonlinear transformations on the features to extract richer feature representation, and use sigmoid as activation function to perform sentiment classification.

7) OUTPUT LAYER

Calculate the probability for each class and select the class with the highest probability as the final prediction sentiment tendency.

B. MECHANISM OF OPERATION

1) INTEGRATION MECHANISM OF BBA

Bert's input is a set of token sequences, in which each token is converted into a vector representation through an embedding layer and then input into the multi-layer encoder of Transformer. BERT's multi-layer encoder performs multi-level feature extraction on the input token sequence to obtain context-related representations of each token. The output of BERT's token-level representation is used as the input of BiLSTM, which further performs context modeling on the text

representation extracted by BERT to obtain a richer sentence-level representation. Then the Attention mechanism further dynamically adjusts the importance of different positions in the sequence obtained by BiLSTM to better capture the semantic information of the sequence.

2) INPUT DATA PROCESSING

Firstly, the course review texts are cleaned, and then the corresponding pre-trained BERT model and its associated tokenizer are loaded based on the specified model name. The loaded pre-trained model files are used to initialize the tokenizer, which performs tokenization, adds special tokens, padding, truncation, and encoding to convert the text into the required input format for the BERT model. The review text is segmented into tokens using a tokenizer, dividing the sentences within the review text. Each word in the review is encoded, and each word is converted into a word vector token embedding, which is added to segment embedding and position embedding, forming the input vector $I = (I_1, I_2, \ldots, I_n)$ of the BERT.

3) BIDIRECTIONAL ENCODER REPRESENTATIONS FROM TRANSFORMERS

BERT is a pre-trained language representation model proposed by Devlin et al. in October 2018 [39]. It is implemented based on transformers and is composed of several stacked transformer encoders. Each transformer encoder consists of a self-attention layer and a feedforward neural network. BERT use two methods of unique Masked Language Model to generate in-depth bi-directional linguistic representations and Next Sentence Prediction that will accurately represent the "word" and "sentence" levels, which makes it powerful and has advantages such as ease of use and strong stability. Here we use a pretrained representation model called "Bert-based-Chinese". It has 12 transformer layers, 768-hidden, 12 multi-head attention, and 110M parameters.

Each self-attention is obtained by the following formula (1)-(4).

$$Q = IW^Q \tag{1}$$

$$K = IW^K \tag{2}$$

$$V = IW^V \tag{3}$$

$$Attention\,(Q, K, V) = softmax\left(\frac{QK^T}{\sqrt{d_k}}\right)V \tag{4}$$

where Q represents the query vector, K represents the key vector, and V represents the numerical vector. These three vectors are obtained from the input vector I through linear transformation. W^Q, W^K, W^V are three trainable parameter matrices. d_k is the dimension size of K.

In multi-head attention, self-attention is computed in parallel multiple times. In this mechanism, the input sequence passes through multiple distinct self-attention sub-layers, each with its own set of weight matrices, to generate different attention representations. Subsequently, these diverse attention representations are combined through linear transformations and concatenation operations to obtain the final multi-head attention representation. The related formulas (5)-(9) are as follows.

$$Q_i = QW_i^Q \tag{5}$$

$$K_i = KW_i^K \tag{6}$$

$$V_i = VW_i^V \tag{7}$$

$$Z_i = Attention\,(Q_i, K_i, V_i) \tag{8}$$

$$Z = Concat\,(Z_1, \cdots, Z_h)\,W^O \tag{9}$$

Then add the residual and apply layer normalization to obtain Z'.

$$Z' = LayerNorm\,(Z + I) \tag{10}$$

It passes through the Feed-Forward Network, which consists of two fully connected layers with a ReLU activation function applied between them. The corresponding formula is:

$$O = ReLU\left(Z'W_1 + b_1\right)W_2 + b_2 \tag{11}$$

Then add the residual and apply normalization once again. The formula is:

$$O' = LayerNorm\left(O + Z'\right) \tag{12}$$

After stacking multiple layers of transformer encoders, the semantic feature vector X of the BERT layer is obtained. The formula is:

$$X = TrsFm_{encoder}\left(O'\right) \tag{13}$$

4) BIDIRECTIONAL LONG SHORT-TERM MEMORY

Long short-term memory network (LSTM) was first proposed by Hochreiter and Schmidhuber in 1997 [40]. It was a kind of temporal recurrent neural network which was better at capturing longer distant dependencies. The structure of the LSTM is illustrated in Fig. 2.

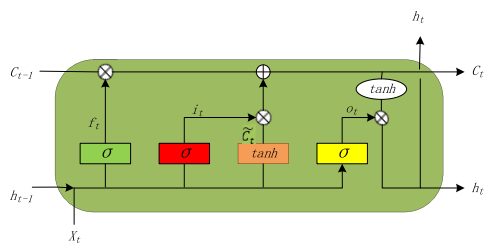

FIGURE 2. Structure of LSTM.

The relevant formulas (14)-(19) are provided below:

$$f_t = \sigma\left(W_f \cdot [h_{t-1}, x_t] + b_f\right) \tag{14}$$

$$i_t = \sigma\left(W_i \cdot [h_{t-1}, x_t] + b_i\right) \tag{15}$$

$$\tilde{C}_t = \tanh\left(W_C \cdot [h_{t-1}, x_t] + b_C\right) \tag{16}$$

$$C_t = f_t * C_{t-1} + i_t * \tilde{C}_t \tag{17}$$

$$O_t = \sigma\left(W_o\,[h_{t-1}, x_t] + b_o\right) \tag{18}$$

$$h_t = O_t * tanh(C_t) \tag{19}$$

where σ denotes a sigmoid neural network structure, and x_t is the input vector at time step t.

Through training, LSTMs learn what information to retain and what to disregard. For example,
"我不觉得他这个人细心." (I don't think he is careful.) The following "细心" is negated by the word "不", that is, the sentence's emotional polarity is derogatory. But one problem with sentence modeling with LSTM is that information cannot be encoded from back to front. For instance, " 这个音响效果差得不行，没有上次上课的效果好"(This sound system has a terrible effect, it's not as good as the last class). Here, the word "不行" refers to a degree alteration of "差".

BiLSTM [41] can better capture bidirectional semantic dependencies. The output from the Bert layer is fed into BiLSTM networks, one running forward and the other running backward. Subsequently, the hidden layers of both networks are concatenated and jointly fed into the attention layer.

Assuming that the feature vectors obtained from the two directions of BiLSTM are h_l and h_r, respectively, the concatenated vector h_i is obtained by combining the vectors from both directions.

$$h_i = concat(h_l, h_r) \tag{20}$$

5) ATTENTION MACHENISM

Attention mechanism can allocate corresponding weights based on the varying importance of words. Adding attention mechanism after the BiLSTM layer allows for a better concentration of attention on the relatively important hidden state information across the entire text paragraph. Assuming $H = [h_1, h_2, \ldots, h_N]$ represents N inputs, the calculation formula for the attention mechanism is as follows:

$$u_i = tanh(W_w h_i + b_w) \tag{21}$$

$$a_i = e^{u_i} \Big/ \sum_j e^{u_j} \tag{22}$$

$$h = \sum_i a_i h_i \tag{23}$$

where the attention distribution a_i represents the degree of correlation between the i-th information in input information vector H. h represents the hidden layer vector outputted by the attention layer.

6) SENTIMENT ANALYSIS OUTPUT

The output of the attention layer enters the pooling layer and the fully connected layer in sequence. Here the activation function we use for the fully connected layer is sigmoid. Assuming the output of the previous layer is f, then we have:

$$\hat{p} = sigmoid\ (Wf + b) \tag{24}$$

$$y = argmax\ (\hat{p}) \tag{25}$$

where $\hat{p}$ represents the output probability for the emotional category, while y is the final emotional output result.

TABLE 2. Data information.

	Negative	Positive
Number	4075	5869
Average length	26	26.3
Number of less than 50 in length	3613	5190

IV. EXPERIMENT

A. DATA SET

Python is used to scrape the course review data of computer science from the Chinese University MOOC platform. We removed duplicates and checked labels, and finally screened 9944 reviews from the online courses, including 4075 negative comments and 5869 positive comments. As is shown in Fig. 3. The description of dataset is shown in Table 2. From Table 2, we can see average length of negative comments is 26 and average length of positive comment is 26.3. The number of negative comments with length less than 50 is 3613, accounting for about 89%. The number of positive comments with length less than 50 is 5190, accounting for about 88%. As is shown in Fig. 4 and 5. So the comments of online courses are mostly short texts. Partial data sample is shown in Table 3.

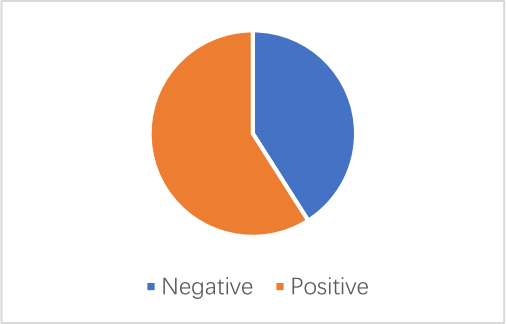

FIGURE 3. Distribution of data.

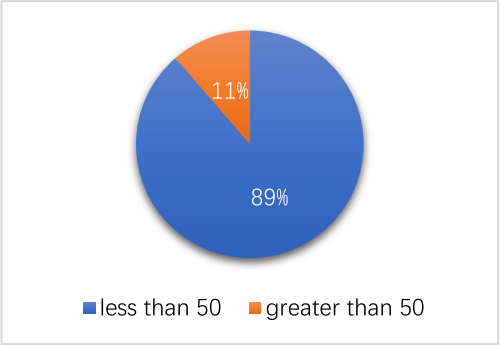

FIGURE 4. Proportion of negative review with length.

B. DATA PROCESSING

Because the course reviews contain a lot of dirty data such as wrong characters and other special symbols, the data is first cleaned to remove these marks.

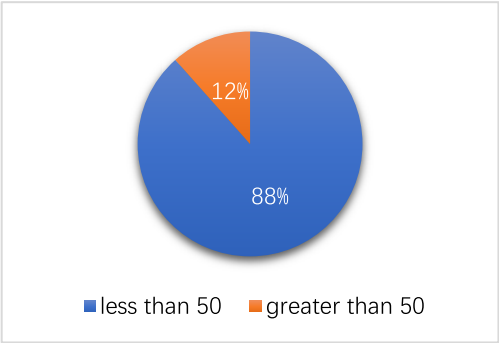

FIGURE 5. Proportion of positive review with length.

TABLE 3. Partial data sample.

sentiment	text
negative	能看得懂的东西，讲得听不懂。(Things that can be understood by reading are incomprehensible when explained.)
positive	编程小白只能说"五星好评"。(A programming novice can only say "five-star praise".)
negative	并没有用，全是理论的东西，根本做不出什么实际的东西来！(It's of no use; it's all theoretical and doesn't lead to anything practical.)
positive	比较细致，比较好。(Comparatively meticulous, comparatively good.)
positive	非常好，尤其网站配置资源，非常丰富，收获很大。(Excellent, especially the website's resource configuration, which is extremely abundant. I gained a lot from it.)
negative	不好，效果不行。(Not good, the results are not satisfactory.)
positive	不愧是国家精品课程，讲得清晰易懂，零基础也能听得懂，课程中的案例看似简单，但是涉及的知识点不少，可以看出老师们非常用心！(It truly deserves to be a national premium course. The explanations are clear and easy to understand, even for beginners. The examples in the course may seem simple, but they cover a lot of knowledge points. It's evident that the teachers put a lot of effort into it!)
negative	不实用，讲得也不仔细。(Not practical, and the explanations are not detailed enough.)
negative	测试题描述模糊，难以下手。(The test questions are described ambiguously, making it difficult to proceed.)
negative	把简单的问题都复杂化了，内容多，更抽象了。(It has overcomplicated simple problems, with more content and greater abstraction.)
positive	把无趣的内容变得很有趣，点赞。(Make boring content interesting. Give it a thumbs up.)
positive	本课程非常值得学习，老师讲得详细，学到了在其它地方没有学习到的内容。(This course is well worth studying. The teacher explains things in detail, and I've learned content that I haven't encountered elsewhere.)
positive	比较灵活实用，让我了解了一些以前不曾知道的与计算机有关的知识。(It is relatively flexible and practical, and it helped me learn some computer-related knowledge that I didn't know before.)

Next, the pre-trained BERT tokenizer is loaded, and the text is further tokenized, special tokens are added, padding/truncation is applied, and it is converted into TensorFlow tensor format, followed by further processing to convert it into the input format required by the BERT model.

C. EXPERIMENT SETUP

After preprocessing, the data is divided into training set, validation set and test set in a ratio of 8:1:1. 7955 samples as training data, 994 samples as validation set, and 995 samples as test set. We use a pretrained representation model called "Bert-based-Chinese". Use BertTokenizer to tokenize the text, and then feed it into the TFBERT pre-trained model. The optimizer we use for the model is Adam. The performance is evaluated using Accuracy, Precision, Recall and F1 value. The corresponding formula is as follows:

- *Accuracy:* $Accuracy = (TP + TN)/(TP + FN + FP + TN)$;
- *Precision:* $Precision = TP/(TP + FP)$;
- *Recall:* $Recall = TP/(TP + FN)$;
- *F1 value:* $F1 = 2 \times Precision \times Recall/(Precision + Recall)$;

Where TP, TN, FP and FN respectively represent the positive samples are projected by the model to be in the positive class, negative samples are anticipated to be in the negative class, the negative samples are anticipated to be in the positive class and positive samples are projected by the model to be in the negative class.

D. EXPERIMENT PROCESS

We use python language to complete experiment based on TensorFlow deep learning framework. After processing, the data scraped from Chinese MOOC platform is divided into train set, validation set and test set. The train set is used to train the model, and the validation set is used for multiple round hyperparameter optimization to improve the model's ability to generalize on unknown data. After training, the test set is used to evaluate the performance of the model. Finally, the trained model is used for sentiment classification. Accuracy, Precision, Recall and F1 value are used as evaluation

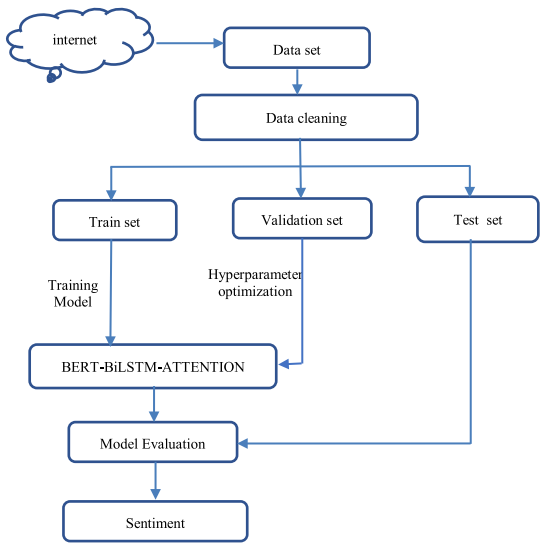

FIGURE 6. Experiment process flow.

metrics in experiment. The whole research process flow is shown in Fig. 6.

TABLE 4. Related parameter settings.

Batch_size:	32
Learning rate:	1e-7
Max_Length:	128
Optimizer:	Adam
Loss function:	binary_crossentropy
Num_class:	2
Epochs:	20

E. EXPERIMENT RESULTS AND ANALYSIS

1) COMPARATION OF LOSS AND ACCURACY

We used 9944 course review data to experiment on our proposed Bert-BiLSTM-Attention. 7955 samples are used to train BBA model, 994 are used to optimize hyperparameters for improving the generalization of the model. 995 samples are used to evaluate the performance of the model. Our BBA model is compared with other models such as BERT-CNN, BERT-LSTM, BERT-CNN-LSTM and BERT-BILSTM. Hyperparameter tuning aims to balance the optimal solution with the regularization term. The model's performance is particularly sensitive to hyperparameters, especially the learning rate, which is a crucial hyperparameter that controls the magnitude of weight updates during training. For the learning rate, we use manual settings based on experience, trying values from high to low. The corresponding parameter optimal settings are shown in Table 4. The batch size is set to 32, the learning rate is 1e-7 and the Max_length is 128. During the training process, the Adam optimization algorithm is used. The loss function used to compute the model error is binary_crossentropy. The evaluation metric used to monitor the model performance during training is accuracy. After 20 epochs of training, the loss value of our BBA model reached 0.1539, with an accuracy of 0.9608. The change of loss and accuracy values of the five models are shown in Fig. 7. We compute the confusion matrix on the test data. The confusion matrix of our model is shown in Fig. 8. We can see from Fig. 8 that there are 956 samples in TP (True Positive) and TN (True Negative), and 39 samples in FP (False Positive) and FN (False Negative).

We compare our BBA model with BERT-CNN, BERT-LSTM, BERT-CNN-LSTM, and BERT-BiLSTM on Accuracy, Precision, Recall and F1-sore. The result of comparation is shown in Table 5. All five models are based on the BERT pre-trained model, but they differ in structure, employing different neural network architectures to handle sentiment analysis tasks. From the experimental results, it can be observed that as the model structure becomes more complex, performance gradually improves. The BERT-CNN model exhibits relatively lower performance, whereas models incorporating LSTM and BiLSTM show further improvement in performance. Particularly, in our model, significant performance enhancement is observed.

TABLE 5. Comparation with other models.

Models	Accuracy	Precision	Recall	F1-Score
BERT-CNN	0.8975	0.8772	0.9565	0.9151
BERT- LSTM	0.9387	0.9227	0.9757	0.9484
BERT-CNN-LSTM	0.9528	0.9430	**0.9774**	0.9599
BERT- BiLSTM	0.9558	0.9477	**0.9774**	0.9623
BERT-BiLSTM-Attention	**0.9608**	**0.9589**	0.9739	**0.9664**

From Table 5, it can be seen that our model achieves the highest accuracy, precision, and F1-score, followed by BERT-BiLSTM, whereas the BERT-CNN model performs the worst.

The accuracy of our model is as high as 0.9608, precision is 0.9589 and the F1-score value is 0.9664. Our model slightly underperforms compared to BERT-CNN-LSTM and BERT-BiLSTM in terms of Recall.

2) THE IMPACT OF ATTENTION MECHANISM

To illustrate the impact of incorporating the attention mechanism on experimental performance, we compare the BERT-BiLSTM model with our BBA model on the test set, as is shown in Table 6 of the experimental data. From the experimental data, it is evident that the BERT-BiLSTM-Attention model outperforms the BERT-BiLSTM model in terms of accuracy, precision and F1.

Both models involve extracting text features based on BERT and modeling sequences with BiLSTM. The addition of the attention mechanism implies that the model can dynamically adjust weights based on the importance of each word while using BiLSTM to process sequences. This allows the model to better understand the relationships between different parts of the input sequence. Moreover, dynamic weight adjustment means the model can better focus on task-relevant parts, thereby enhancing its ability to capture key information. By adapting weights based on contextual information, the model can also more accurately learn representations from the input sequence. In contrast, the BERT-BiLSTM model may encounter problems of information imbalance

TABLE 6. Impact of attention mechanism.

Models	Accuracy	Precision	Recall	F1-Score
BERT- BiLSTM	0.9558	0.9477	**0.9774**	0.9623
BERT- BiLSTM-Attention	**0.9608**	**0.9589**	0.9739	**0.9664**

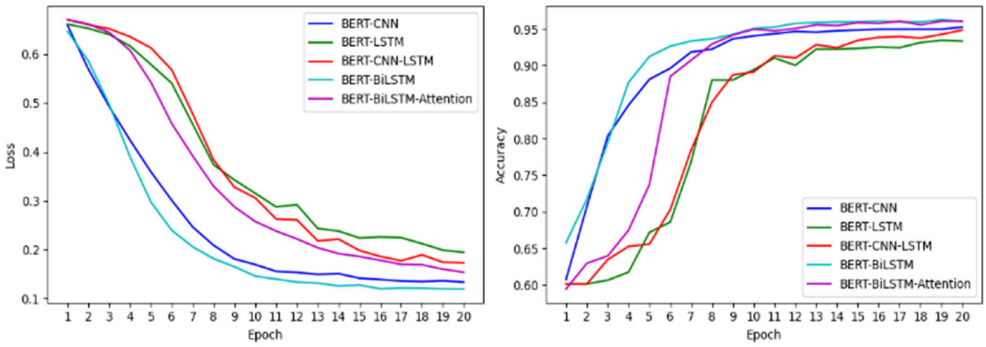

FIGURE 7. The loss and accuracy curve.

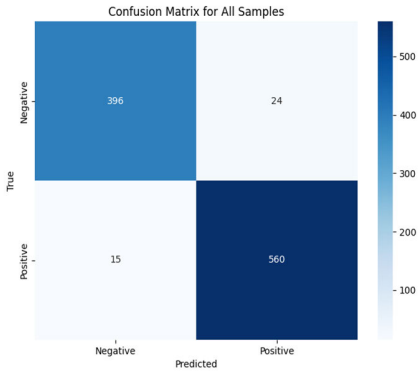

FIGURE 8. Confusion matrix.

when processing sequences and may be more susceptible to interference from specific parts of the sequence.

3) THE IMPACT OF BIDIRECTIONALITY OF BILSTM

To better explore the impact of the bidirectionality of BiLSTM on the model, we conducted a comparison between the BERT-LSTM-Attention model and our model, with the experimental results shown in Table 7. From the experimental results, it can be observed that our model outperforms the BERT-LSTM-Attention model across all four metrics.

Both models integrate BERT for contextual embeddings and attention mechanisms to capture subtle dependencies within sequences. However, they differ in the recurrent layer used for sequence modeling, with BERT-LSTM-Attention employing LSTM (Long Short-Term Memory) and our model utilizing BiLSTM (Bidirectional Long Short-Term Memory). BiLSTM combines information from both past and future states of a sequence, enabling it to effectively capture bidirectional dependencies and thereby model complex relationships and patterns within sequence data. Due to its ability to capture bidirectional context, BiLSTM exhibits greater robustness to semantic ambiguity present in natural language. It can leverage both preceding and succeeding words to disambiguate meanings and make more informed predictions. In contrast, LSTM processes sequences in only one direction, potentially limiting its ability to fully understand the context.

In summary, our model leverages the bidirectional processing capability of BiLSTM, which allows it to effectively capture bidirectional dependencies, model complex sequence patterns, and exhibit greater robustness to semantic ambiguity compared to the unidirectional processing of LSTM in the BERT-LSTM-Attention model. These factors contribute to the superior performance of our model observed in the experimental results.

The experiment shows that our proposed model is effective in sentiment classification of course comments. Teachers can analyze the sentiment data of course comments, adjust teaching content or methods in a timely manner, and improve teaching quality. At the same time, teachers can also understand the learning status of specific students based on their comment information, and provide personalized teaching information for students.

TABLE 7. Impact of bidirectionality of bilstm.

Models	Accuracy	Precision	Recall	F1-Score
BERT-LSTM-Attention	0.9508	0.9534	0.9617	0.9576
BERT-BiLSTM-Attention	**0.9608**	**0.9589**	**0.9739**	**0.9664**

V. CONCLUSION

Our model has achieved excellent results, partly due to the cleaning of the data, ensuring data quality. Additionally, it combines the advantages of BERT, BiLSTM, and the attention mechanism. BERT promotes dynamic feature representation through MASK and Next Sentence Prediction, demonstrating robust generalization capabilities. Furthermore, it seamlessly integrates the bidirectional processing capability of BiLSTM, enabling the model to gain a deeper understanding of word dependencies within text sequences. This advantage is further enhanced by a refined attention mechanism, optimizing the allocation of word weights to comprehensively capture contextual information in text data.

The fusion of these advanced technologies significantly enhances the model's representation learning capability, leading to superior performance in sequence modeling and information capture. Experimental results indicate that applying the BBA deep learning model for sentiment analysis of course evaluations can effectively understand students' emotions and learning experience. There are certain limitations to using text only for sentiment analysis. After all, other modalities, such as audio, video, body movements, etc., also contain rich emotions. Video data can capture changes in facial expressions and body language, and audio can capture the students' tone. The complementarity of multimodal data can reduce the information loss and misjudgment that may be caused by a single text modality. Our research will focus on conducting sentiment analysis of students using multimodal data. More accurate analysis of emotions through multi-modality will provide broad prospects for assisting in improving students' learning abilities and teachers' teaching effectiveness.

REFERENCES

[1] X. Jiang, C. Song, Y. Xu, Y. Li, and Y. Peng, "Research on sentiment classification for netizens based on the BERT-BiLSTM-TextCNN model," *PeerJ Comput. Sci.*, vol. 8, p. e1005, Jun. 2022, doi: 10.7717/peerj-cs.1005.

[2] A. F. Ibrahim, M. Hassaballah, A. A. Ali, and I. A. Ibrahim, "A study of sentiment analysis approaches in short text," in *Digital Transformation Technology*, vol. 224. Singapore: Springer, 2022, pp. 143–151, doi: 10.1007/978-981-16-2275-5_8.

[3] W. Liu, G. Cao, and J. Yin, "Bi-level attention model for sentiment analysis of short texts," *IEEE Access*, vol. 7, pp. 119813–119822, 2019.

[4] J. Wu, T. Y. Liu, and W. Hu, "Document-level sentiment analysis of course review based on BG-caps," in *Proc. Int. Conf. Pioneering Comput. Scientists, Eng. Educators (ICPCSEE)*, Chengdu, China, 2022, pp. 394–405, doi: 10.1007/978-981-19-5209-8_26.

[5] Y. Zhang, Z. Zhang, D. Miao, and J. Wang, "Three-way enhanced convolutional neural networks for sentence-level sentiment classification," *Inf. Sci.*, vol. 477, pp. 55–64, Mar. 2019.

[6] Q. Liu, B. Liang, J. Xu, and Q. Zhou, "A deep hierarchical neural network model for aspect-based sentiment analysis," *Chin. J. Comput.*, vol. 41, no. 12, pp. 2637–2652, 2018.

[7] Z. Kastrati, A. S. Imran, and A. Kurti, "Weakly supervised framework for aspect-based sentiment analysis on students' reviews of MOOCs," *IEEE Access*, vol. 8, pp. 106799–106810, 2020.

[8] M. Ojeda-Hernández, D. López-Rodríguez, and Á. Mora, "Lexicon-based sentiment analysis in texts using formal concept analysis," *Int. J. Approx. Reasoning*, vol. 155, pp. 104–112, Apr. 2023.

[9] C. S. Khoo and S. B. Johnkhan, "Lexicon-based sentiment analysis: Comparative evaluation of six sentiment lexicons," *J. Inf. Sci.*, vol. 44, no. 4, pp. 491–511, Aug. 2018.

[10] G. Xu, Z. Yu, H. Yao, F. Li, Y. Meng, and X. Wu, "Chinese text sentiment analysis based on extended sentiment dictionary," *IEEE Access*, vol. 7, pp. 43749–43762, 2019.

[11] S. Zhang, Z. Wei, Y. Wang, and T. Liao, "Sentiment analysis of Chinese micro-blog text based on extended sentiment dictionary," *Future Gener. Comput. Syst.*, vol. 81, pp. 395–403, Apr. 2018.

[12] X. P. Yang, Z. X. Zhang, L. Wang, Y. J. Zhang, Q. F. Ma, J. N. Wu, and Y. Zhang, "Automatic construction and optimization of sentiment dictionary based on Word2Vec," *Comput. Sci.*, vol. 44, no. 1, pp. 42–47, 2017.

[13] R. Obiedat, R. Qaddoura, A. M. Al-Zoubi, L. Al-Qaisi, O. Harfoushi, M. Alrefai, and H. Faris, "Sentiment analysis of customers' reviews using a hybrid evolutionary SVM-based approach in an imbalanced data distribution," *IEEE Access*, vol. 10, pp. 22260–22273, 2022.

[14] H. Benarafa, M. Benkhalifa, and M. Akhloufi, "WordNet semantic relations based enhancement of KNN model for implicit aspect identification in sentiment analysis," *Int. J. Comput. Intell. Syst.*, vol. 16, no. 1, p. 3, Jan. 2023.

[15] H. Rehioui and A. Idrissi, "New clustering algorithms for Twitter sentiment analysis," *IEEE Syst. J.*, vol. 14, no. 1, pp. 530–537, Mar. 2020.

[16] S. Zhang, "Sentiment classification of news text data using intelligent model," *Frontiers Psychol.*, vol. 12, Sep. 2021, Art. no. 758967, doi: 10.3389/fpsyg.2021.758967.

[17] Y. J. Su, C. H. Chen, T. Y. Chen, and C. C. Cheng, "Chinese microblog sentiment analysis by adding emoticons to attention-based CNN," *J. Internet Technol.*, vol. 21, no. 3, pp. 821–829, 2020.

[18] L.-X. Luo, "Network text sentiment analysis method combining LDA text representation and GRU-CNN," *Pers. Ubiquitous Comput.*, vol. 23, nos. 3–4, pp. 405–412, Jul. 2019.

[19] Z. Gao, Z. Li, J. Luo, and X. Li, "Short text aspect-based sentiment analysis based on CNN + BiGRU," *Appl. Sci.*, vol. 12, no. 5, p. 2707, Mar. 2022, doi: 10.3390/app12052707.

[20] L. Yan, J. Han, Y. Yue, L. Zhang, and Y. Qian, "Sentiment analysis of short texts based on parallel DenseNet," *Comput., Mater. Continua*, vol. 69, no. 1, pp. 51–65, 2021.

[21] J. L. Huan, A. A. Sekh, C. Quek, and D. K. Prasad, "Emotionally charged text classification with deep learning and sentiment semantic," *Neural Comput. Appl.*, vol. 34, no. 3, pp. 2341–2351, Feb. 2022.

[22] Z. He, "Text sentiment analysis based on multi-layer bi-directional LSTM with a trapezoidal structure," *Intell. Autom. Soft Comput.*, vol. 37, no. 1, pp. 639–654, 2023.

[23] H. Liu, X. Chen, and X. Liu, "A study of the application of weight distributing method combining sentiment dictionary and TF-IDF for text sentiment analysis," *IEEE Access*, vol. 10, pp. 32280–32289, 2022.

[24] S. Hao, P. Zhang, S. Liu, and Y. Wang, "Sentiment recognition and analysis method of official document text based on BERT–SVM model," *Neural Comput. Appl.*, vol. 35, no. 35, pp. 24621–24632, Dec. 2023.

[25] M. Kamyab, G. Liu, and M. Adjeisah, "Attention-based CNN and bi-LSTM model based on TF-IDF and GloVe word embedding for sentiment analysis," *Appl. Sci.*, vol. 11, no. 23, p. 11255, Nov. 2021, doi: 10.3390/app112311255.

[26] B. Shin, T. Lee, and J. D. Choi, "Lexicon integrated CNN models with attention for sentiment analysis," 2016, *arXiv:1610.06272*.

[27] J. Khan, N. Ahmad, S. Khalid, F. Ali, and Y. Lee, "Sentiment and context-aware hybrid DNN with attention for text sentiment classification," *IEEE Access*, vol. 11, pp. 28162–28179, 2023.

[28] H. Li, Y. Lu, H. Zhu, and Y. Ma, "A novel AB-CNN model for multi-classification sentiment analysis of e-Commerce comments," *Electronics*, vol. 12, no. 8, p. 1880, Apr. 2023, doi: 10.3390/electronics12081880.

[29] Y. Shuoqiu and X. Chaojun, "Research on constructing sentiment dictionary of online course reviews based on multi-source combination," in *Proc. 2nd Int. Conf. Data Sci. Inf. Technol.*, vol. 33, Jul. 2019, pp. 71–76.

[30] K. Wang and Y. Zhang, "Topic sentiment analysis in online learning community from college students," *J. Data Inf. Sci.*, vol. 5, no. 2, pp. 33–61, Apr. 2020.

[31] X. Yuan, "Emotional tendency of online legal course review texts based on SVM algorithm and network data acquisition," *J. Intell. Fuzzy Syst.*, vol. 37, no. 5, pp. 6253–6263, Nov. 2019.

[32] A. Onan, "Sentiment analysis on massive open online course evaluations: A text mining and deep learning approach," *Comput. Appl. Eng. Educ.*, vol. 29, no. 3, pp. 572–589, May 2021.

[33] X. Li, H. Zhang, Y. Ouyang, X. Zhang, and W. Rong, "A shallow BERT-CNN model for sentiment analysis on MOOCs comments," in *Proc. IEEE Int. Conf. Eng., Technol. Educ. (TALE)*, Yogyakarta, Indonesia, Dec. 2019, pp. 1–6.

[34] C. Qi and S. Liu, "Evaluating on-line courses via reviews mining," *IEEE Access*, vol. 9, pp. 35439–35451, 2021.

[35] S. Ji and T. Fangbi, "Emotion analysis model of MOOC course review based on BiLSTM," *Int. J. Emerg. Technol. Learn. (iJET)*, vol. 16, no. 8, pp. 93–105, Apr. 2021.

[36] X. Zhang and X. Qin, "Research on sentiment analysis algorithm for comments on online ideological and political courses," *Int. J. Adv. Comput. Sci. Appl.*, vol. 13, no. 11, pp. 174–179, 2022.

[37] S. Wang, Q. Feng, and J. Sun, "Deep learning sentiment analysis for MOOC course reviews," in *Advances in Natural Computation, Fuzzy Systems and Knowledge Discovery*, vol. 89. Cham, Switzerland: Springer, 2021, pp. 838–848, doi: 10.1007/978-3-030-89698-0_86.

[38] L. P. Hung and S. Alias, "Beyond sentiment analysis: A review of recent trends in text based sentiment analysis and emotion detection," *J. Adv. Comput. Intell. Intell. Informat.*, vol. 27, no. 1, pp. 84–95, Jan. 2023.

[39] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, "BERT: Pre-training of deep bidirectional transformers for language understanding," 2018, *arXiv:1810.04805*.

[40] S. Hochreiter and J. Schmidhuber, "Long short-term memory," *Neural Comput.*, vol. 9, no. 8, pp. 1735–1780, Nov. 1997, doi: 10.1162/neco.1997.9.8.1735.

[41] A. Graves and J. Schmidhuber, "Framewise phoneme classification with bidirectional LSTM and other neural network architectures," *Neural Netw.*, vol. 18, nos. 5–6, pp. 602–610, Jul. 2005.

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